

- A small switch- push button kind: This will indicate if your Arduino is turned on or not.
- Decoupling capacitors of electrolytic type
- A voltage regulator of 5V
- One crystal of 16MHz
- A ceramic capacitor of 22pf
- Female header
- Standoffs

For programming

- FT232RL USB to Serial: You can use other kinds but this one seems to work the best.
- A breadboard: This will be the final base for all your components, hardware and software alike.
- Wires- both short and long
- An LED or a bar graph display and potentiometer.

List of tools required

- A Dremel: A tool used for rotating purposes, you will use a Dremel to drill holes. A hacksaw can also act as a substitute
- 1/32" Drill bit: Dremel can be used here, too.
- A disc that cuts metals: You will use this to cut the copper board
- Solder
- Desoldering braid
- Soldering gripper
- Flux
- Soldering iron

2. Making your own printed circuit board or PCB

This step involves the overall procedure of building an Arduino. It is advised that the reader goes through the entire article before following this procedure.

PCB or printed circuit boards are the primary support systems that form the mechanical backbone of electrical components that are used for connecting some other electrical parts. These are usually used with copper sheets since they provide the best results. Also, copper is the most commonly-used metal if you are trying to build something like the Arduino.

To make it clear, you can easily buy a PCB from any hardware shop. However, there is an advantage of using one that you have made. The PCB